MEGATHON X

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QUALCOMM PROBLEM STATEMENT

AI Incident Report Assistant

End-to-end voice-driven incident reporting using on-device Generative AI and function calling.

Qualcomm

ON-DEVICE AI CHALLENGE

Voice in. Structured report out.

IN COLLABORATION WITH

INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY HYDERABAD

E-Cell, IIIT Hyderabad

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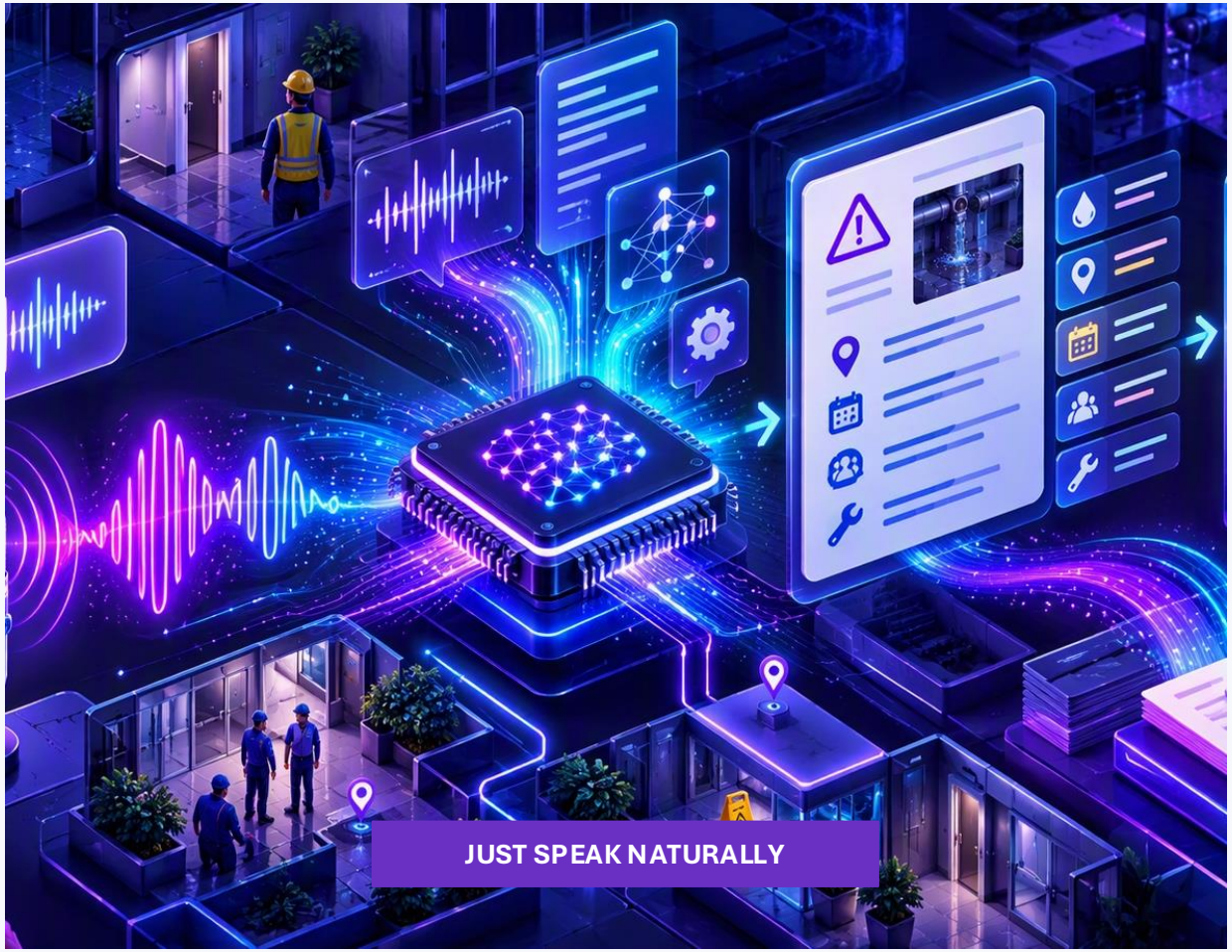
How many of you enjoy filling out forms?

Let us replace manual entry with a natural conversation.

Probably not many.

Imagine an incident happens. Instead of opening a portal and typing every detail, you simply talk to the device.

> *“There was a water leak near Conference Room 2 around 10 AM. Maintenance has already isolated the area.”*

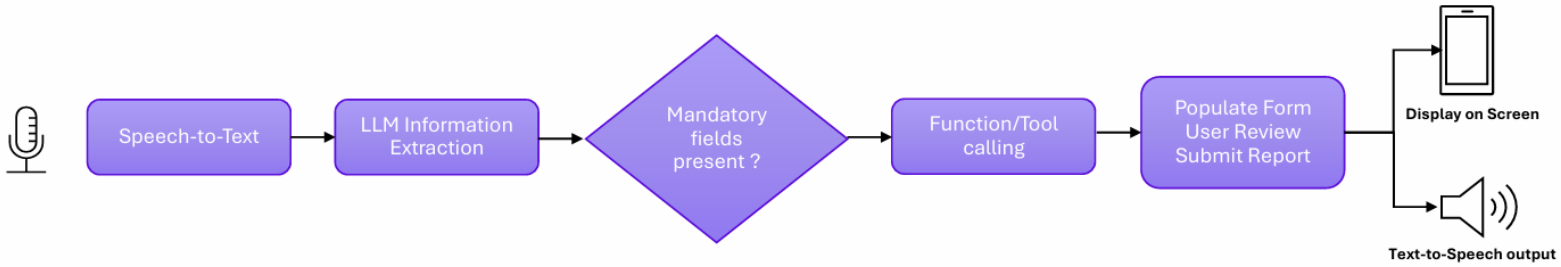


Problem Statement

AI Incident Report Assistant

Problem Statement 1 – AI Incident Report Assistant

End-to-end voice-driven incident reporting using on-device Generative AI and function calling.



Example Fields of a Form

The application should automatically populate fields such as:

- Incident Type
- Severity Level
- Incident Location
- Date & Time
- Persons Involved
- Equipment / Asset Involved
- Incident Description
- Immediate Actions Taken
- Witness Information
- Recommended Follow-Up Action

Example Input

"There was a water leak near Conference Room 2 around 10 AM. Maintenance has already isolated the area."

Example Output

- Incident Type → Water Leak
- Location → Conference Room 2
- Time → 10 AM
- Action Taken → Area Isolated

Evaluation Criteria

- All AI models must execute on-device using Snapdragon acceleration. Strictly no cloud APIs.
- Ability to understand natural conversational speech rather than predefined commands.
- Accuracy of speech recognition and transcription.
- Accuracy of extracting incident information from user narration.
- Correct mapping of extracted information to report fields.

Build with the Qualcomm® Innovators Development Kit (QIDK)

Start with these resources for models, SDK setup, and QIDK development.

01

QIDK Documentation

Release notes and platform overview

Open resource →

02

QIDK GitHub Repository

Source code, examples, and project resources

Open resource →

03

Qualcomm AI Hub

Models compatible with Snapdragon mobile chipsets

Open resource →

04

QAIRT SDK Setup Guide

Environment and SDK setup documentation

Open resource →

YOUR CHALLENGE

Make incident reporting as easy as speaking.

Natural conversation • Accurate extraction • On-device execution

SPEAK → UNDERSTAND → EXTRACT → REVIEW → SUBMIT

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Thank you

We look forward to your ideas.

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